

Worksheet 4

Problem 1. Linear Functionals. Let $v_1, \dots, v_n$ be a basis for a real vector space V , and let $\varphi_1, \dots, \varphi_n$ be associated dual basis. Fix some integer $0 < m \leq n$, and define a function $T : V \rightarrow \mathbb{R}^m$ by the rule

$$T(v) = \begin{bmatrix} \varphi_1(v) \\ \varphi_2(v) \\ \vdots \\ \varphi_m(v) \end{bmatrix}.$$

- (a) Prove that T is a linear transformation.
- (b) Determine the rank of T .
- (c) Use part (b) to quickly find a basis for the nullspace of T .

Problem 2. Quotient Spaces. Let $V = \mathcal{P}_3(\mathbb{R})$ be the space of real cubic polynomials. Define a linear transformation $T : V \rightarrow V$ by the rule

$$T(p) = p''.$$

That is, T is the linear transformation sending a polynomial to its second derivative.

- (a) Check that the subspace
$$U = \text{span}(1, x)$$
is the nullspace of T .
- (b) Describe the quotient space V/U . What do vectors in this space look like? What does addition look like? What is its dimension?
- (c) Prove that $\hat{T} : V/U \rightarrow V/U$ is the zero transformation, that is, it sends everything to the zero vector.

Problem 3. Linear Operators. Let $T : V \rightarrow V$ be a linear transformation from some vector space V back to itself. We define the *square* $T^2 : V \rightarrow V$ as the function $T^2(v) = T(T(v))$.

- (a) Prove that T^2 is a linear function.
- (b) Show that $\text{null}(T) \subset \text{null}(T^2)$.
- (c) Show that $\text{range}(T^2) \subset \text{range}(T)$.